



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name : Atharva Ambilwade

Class : CSE

PRN : 25070122055

Batch : 2025-29

Subject: Programming with Java

ASSIGNMENT No : 10

TITLE : Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.



Code and Output Screenshots

```
J Main9.java > Java > Main9
1  abstract class Shape {
2      abstract void draw();
3
4      void message () {
5          System.out.println("This is a shape");
6      }
7  }
8
9  class Circle extends Shape {
10     void draw() {
11         System.out.println("This call draws a circle.");
12     }
13 }
14
15 public class Main9 {
16     Run main | Debug main | Run | Debug
17     public static void main (String[] args) {
18         Shape s = new Circle();
19         s.draw();
20         s.message();
21     }
22 }
```

PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main9.java && java Main9
This call draws a circle.
This is a shape

[Done] exited with code=0 in 0.494 seconds
```

Conclusion

This program demonstrates the use of abstract classes and abstract methods in Java. An abstract class provides a common structure, while abstract methods are implemented by derived classes according to their specific requirements. This improves code reusability, flexibility, and supports abstraction in object-oriented programming.



Exercise

1. Create an abstract Payment class and implement Credit Card and UPI payment classes.?

```
J Main1.java > Java > Main1
1  abstract class Payment {
2      |   abstract void pay();
3  }
4  class CreditCard extends Payment {
5      |   void pay() {
6      |       |   System.out.println("Payment made using Credit Card.");
7      |       }
8  }
9  class UPI extends Payment {
10     |   void pay() {
11     |       |   System.out.println("Payment made using UPI.");
12     |       }
13 }
14 public class Main1 {
15     |   Run main | Debug main | Run | Debug
16     |   public static void main(String[] args) {
17     |       |   CreditCard c = new CreditCard();
18     |       |   UPI u = new UPI();
19     |       |   c.pay();
20     |       |   u.pay();
21 }
21 }
```

PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, lexcludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main1.java && java Main1

Payment made using Credit Card.

Payment made using UPI.

[Done] exited with code=0 in 0.504 seconds



2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
J Main7.java > Java > Main7
1  abstract class FoodOrder {
2      |   abstract void calculateBill();
3  }
4  class DineInOrder extends FoodOrder {
5      |   void calculateBill() {
6      |       |   System.out.println("Dine-In Bill: Rs. 5000");
7      |       |   }
8      |   }
9  class TakeAwayOrder extends FoodOrder {
10     |   void calculateBill() {
11     |       |   System.out.println("Take Away Bill: Rs. 4650");
12     |       |   }
13     |   }
14  public class Main7 {
15     |   Run main | Debug main | Run | Debug
16     |   public static void main(String[] args) {
17     |       |   DineInOrder d = new DineInOrder();
18     |       |   TakeAwayOrder t = new TakeAwayOrder();
19     |       |
20     |       |   d.calculateBill();
21     |       |   t.calculateBill();
22     |       |   }
23     |   }
```

PROBLEMS 21 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, !excludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main7.java && java Main7

Dine-In Bill: Rs. 5000

Take Away Bill: Rs. 4650

[Done] exited with code=0 in 0.507 seconds